Lesson plan · for the teacher

From words to a model

Optimization · Level 2 · Lesson 1

Time
40 min

Materials
Board and chalk; A price list for exercise books and pens (made up or from a local shop); Paper for each group

Learning goals

- Write decisions as variables with units.
- Write limits as inequalities and the goal as one expression.
- Check a plan by putting its numbers in.

Activity: Model a class purchase

1. Warm-up (5 min): Ask what the class needs for the term and write the answers. Choose two items, such as exercise books and pens, and write their prices and the budget.
2. Teach (10 min): Write three headings: variables (what we decide, with units), constraints (limits as ≥ or ≤) and objective (one expression to make big or small). Show the school lunch as an example.
3. Model (10 min): With the class, write the variables (b exercise books, p pens), the constraints (cost within the budget, at least so many of each) and the goal, such as the most items, or money left for the class box.
4. Check (10 min): Each group proposes a plan. The class puts its numbers into every constraint and marks it feasible or not. Which feasible plan is best for the goal?
5. Wrap-up (5 min): Ask what the model leaves out (pupils who can't buy their own, colours, quality). Agree one rule you would add.

Use small, round prices so pupils can check plans in their heads, for example books at 8 coins, pens at 5 and a budget of 200.

Discussion

- Why are prices data and not variables?
- Why does "nearly" not count for a constraint?
- Which rule did our model forget, and who would notice first?

Exit ticket

- Write one variable, one constraint and the objective for a decision at home.

Try it at school tomorrow
Model a class purchase with your pupils: exercise books and pens, a budget, and how many of each the class needs. Write the variables, the limits and the goal on the board, then check each group's plan by putting in its numbers. Finish by asking what the model is missing.

Practice questions · for you

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1 **Mr Okafor's lunch model has the variables and every limit (energy, protein, iron), but he forgot the goal. What will a solver give him?**

◯ The cheapest plate ◯ Any plate that keeps the limits, maybe a costly one

◯ An error: no plate exists

2 **The school garden: x tomato beds and y bean beds. Match each sentence to its maths.**

Land: at most 8 beds in all	• • • •	x + 3y ≤ 18
Water: 2 cans a day per tomato bed, 1 per bean bed, at most 14	• • • •	x + y ≤ 8
Hours: 1 a week per tomato bed, 3 per bean bed, at most 18	• • • •	30x + 40y
Profit: 30 coins a tomato bed, 40 a bean bed	• • • •	2x + y ≤ 14

3 **This program checks a plate against the energy limit. What does it print?**

```
rice = 2
nuts = 1
energy = 260 * rice + 170 * nuts
print(energy >= 700)
```

◯ False ◯ 690 ◯ True

4 **Which of these are good variables for the lunch model? Choose all that apply.**

☐ Cups of greens per pupil ☐ Whether lunch is tasty ☐ Handfuls of groundnuts per pupil

☐ The price of a cup of rice ☐ Cups of rice per pupil

5 **A plate that misses the iron limit by only a little is still feasible.**

◯ True ◯ False

✎ **Pick a decision at your school that uses numbers: a budget, a rota, a timetable. What would its variables, constraints and objective be, and what rule might nobody think to write down?**

Check your answers

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1 **Any plate that keeps the limits, maybe a costly one**

Without an objective, every feasible plate is an equally good answer, so the solver returns one that keeps the limits, with no reason to make it cheap.

2 **Land: at most 8 beds in all — $x + y \leq 8$**
Water: 2 cans a day per tomato bed, 1 per bean bed, at most 14 — $2x + y \leq 14$
Hours: 1 a week per tomato bed, 3 per bean bed, at most 18 — $x + 3y \leq 18$
Profit: 30 coins a tomato bed, 40 a bean bed — $30x + 40y$

Each limit adds up what every bed uses and caps the total: land counts beds, water counts cans, hours count pupils' time.

3 **False**

$260 \times 2 + 170 \times 1 = 690$, and $690 \geq 700$ is not true, so it prints False: the plate breaks the energy limit by 10 kcal.

4 **Cups of rice per pupil; Handfuls of groundnuts per pupil; Cups of greens per pupil**

Portions are decisions with a unit, so they are variables.

5 **False**

Feasible means every constraint holds. A plate with 4.9 mg of iron breaks "iron ≥ 5".